

Article

Electrical Impedance Tomography Reconstruction with Partial Boundary Measurements

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Abstract

Electrical Impedance Tomography (EIT) is a non-invasive imaging technique that enables real-time monitoring of human organ function by reconstructing the internal conductivity distribution. In clinical settings, electrode placement is often constrained, resulting in incomplete boundary measurements and a consequent severe degradation of image reconstruction quality. In this work, we investigate EIT reconstruction under partially covered electrode boundaries using our recently proposed nonlinear weighted anisotropic total variation (NWATV) regularizer. Through numerical simulations and water tank experiments with three-quarter and half-circumferential electrode configurations, we demonstrate that NWATV significantly improves both structural fidelity and robustness in reconstructed images under partial electrode coverage. Our findings indicate that NWATV holds potential for clinical use in limited-electrode configurations.

Keywords: EIT; partial boundary data; regularization; NWATV

MSC: 35R30; 65N21; 78A46; 92C55

1. Introduction

Electrical Impedance Tomography (EIT) is an imaging modality that visualizes the internal structure of the imaging object Ω through reconstructing the admittivity distribution $\gamma = \sigma + i\omega\epsilon$ from the boundary electrical measurement. Here, σ denotes the conductivity distribution, ϵ denotes the permittivity distribution, i denotes the imaginary unit, and ω denotes the angular frequency.

In an E -channel EIT system, we inject a sinusoidal current $i(t) = I \sin \omega t$ at the low frequency $\omega/2\pi \leq$ of 10 MHz to the object $\Omega \subset \mathbb{R}^n$ ($n = 2, 3$) through the surface electrodes $\mathcal{E}_j$ ($j = 1, 2, \dots, E$) attached on $\partial\Omega$, the boundary of Ω . The injected current will induce an electrical field $\mathbf{E} = -\nabla u$ inside the object, where u represents the resulting voltage potential. For a current applied between an electrode pair $(\mathcal{E}_j, \mathcal{E}_{j+1})$, the corresponding potential u^j satisfies the following elliptic partial differential equation (PDE) with mixed boundary conditions (Cheney et al. [1]).



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$$\begin{cases} \nabla \cdot (\sigma \nabla u^j) = 0, & \text{in } \Omega \\ \sigma \nabla u^j \cdot \mathbf{n} = 0, & \text{on } \partial\Omega \setminus \bigcup_{i=1}^E \mathcal{E}_i \\ \int_{\mathcal{E}_j} \sigma \frac{\partial u^j}{\partial \mathbf{n}} dS = I = - \int_{\mathcal{E}_{j+1}} \sigma \frac{\partial u^j}{\partial \mathbf{n}} dS, \\ u^j + z_i \sigma \frac{\partial u^j}{\partial \mathbf{n}} \Big|_{\mathcal{E}_i} = U_i^j, & i, j = 1, 2, \dots, E \\ \int_{\mathcal{E}_k} \sigma \frac{\partial u^j}{\partial \mathbf{n}} dS = 0, & k \in \{1, 2, \dots, E\} \setminus \{j, j+1\}. \end{cases} \quad (1)$$

Here, z_i denotes the contact impedance between $\partial\Omega$ and $\mathcal{E}_i$, $\mathbf{n}$ is the unit outer normal vector on $\partial\Omega$, and U_i^j denotes the voltage potential on $\mathcal{E}_i$ resulting from the current injection via the pair $(\mathcal{E}_j, \mathcal{E}_{j+1})$ (Cheng et al. [2]). Using an EIT scanner, one can measure the boundary voltage difference $V_i^j := U_i^j - U_{i+1}^j$ for $i \in \{1, 2, \dots, E\}$, where we set $E+1$ to be 1.

The presence of conductivity or permittivity contrasts perturbs the direction of electrical streamlines, which in turn induces measurable changes in the boundary voltages U_i^j . Mathematically, EIT can be formulated as an inverse problem aimed at reconstructing the admittivity distribution γ from measured voltage differences (see Figure 1). EIT finds broad applications in biomedical engineering, geophysical prospection, and industrial process monitoring. Its advantages are that it is noninvasive, portable, low-cost, radiation-free, and capable of long-term continuous monitoring with high temporal resolutions, all of which make it particularly suitable for bedside pulmonary function testing in both home care and intensive care unit (ICU) settings.

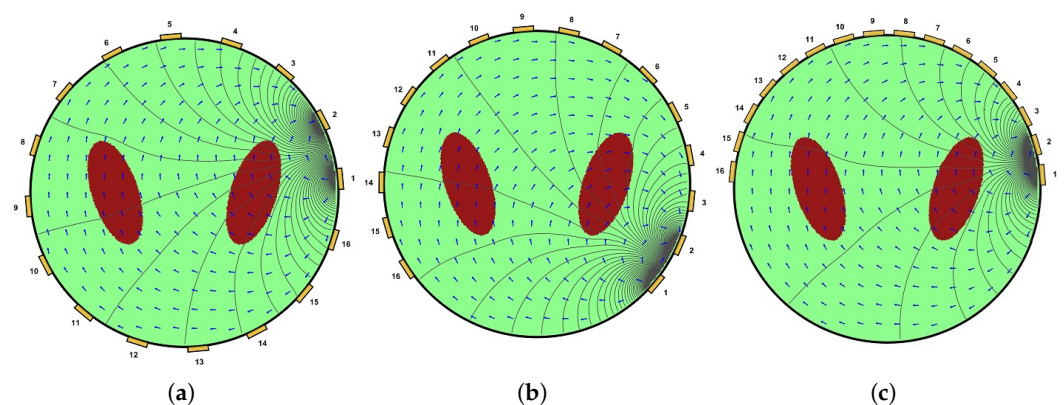


Figure 1. Configuration of a 16-channel EIT system with electrodes evenly spaced (a) along the entire boundary, (b) along three-quarters of the boundary, and (c) along half of the boundary. In this figure, the blue arrows indicate the direction of current flow, and the solid lines within the domain represent equipotential lines. The colors red and green are used to distinguish between conductivities of 2 S/m and 1 S/m, respectively.

However, conventional EIT systems require electrodes to be evenly distributed over the entire boundary $\partial\Omega$, which can cause discomfort in patients, especially in the back. Furthermore, this full-boundary arrangement is highly susceptible to electrode displacement. Since the boundary voltage U_i^j is often more sensitive to such displacement than to local conductivity variations away from the measuring electrodes, even slight movements can introduce severe artifacts in the reconstructed images [3]. Therefore, a technology that operates with electrodes attached only to a portion of the boundary, for instance covering only three-quarters or half of it, would be highly desirable (see Figure 1b,c).

The uniqueness of EIT reconstruction from partial boundary measurements is established in several theoretical works, such as those by [4–6]. Following these foundational

contributions, numerous reconstruction algorithms have been proposed, which can generally be classified into the following categories: (i) direct reconstruction methods, e.g., the d-bar-based approach introduced by [7]; (ii) methods that reconstruct using approximated full-boundary data obtained by solving an optimization problem on the coefficients of the Neumann-to-Dirichlet map [8]; (iii) network-based methods, including the circular and pyramidal resistor network techniques proposed by [9,10], which are designed to compute solutions on optimal grids; (iv) sparsity-constrained least squares approaches, as presented in [11]; (v) complex geometrical optics methods for 3D EIT reconstruction [12]. While the above methods have demonstrated effectiveness in simulation studies, to our knowledge, except (v), none have yet been validated on real EIT measurement data. Moreover, although (v) exhibit fast computation times, their performance, as indicated by experimental studies, still falls behind that of total-variation-regularized least squares methods.

In this work, we apply our previously proposed nonlinear weighted anisotropic total variation (NWATV) regularization strategy [13], originally developed for full boundary EIT reconstruction, to the partial boundary measurement setting, aiming to recover the conductivity distribution under limited electrode coverage. Results show that the NWATV approach can reconstruct internal structures more accurately and with improved computational efficiency. We validate our findings using both numerical simulations and experimental data. The proposed method is compared against conventional regularization strategies, including the fidelity embedding regularizer [14], standard total variation regularizers [15–17], and the first-order TV regularizer [18]. Quantitative evaluation based on the peak signal-to-noise ratio (PSNR), relative error (RE), and CPU time demonstrates that NWATV offers a clear advantage in preserving edge information and maintaining reconstruction fidelity, particularly when electrode coverage is below 3-quarters. Moreover, compared to state-of-the-art methods, the contributions of this work, detailed below, make NWATV a promising candidate for clinical applications where electrode placement is restricted:

- The regularization is adaptive in the sense that, with the same selected regularization parameter, the internal edges and smooth regions diffuse to different degrees;
- The reconstruction is near-real-time, which makes it useful for monitoring the lung in the intensive care unit (ICU);
- The proposed algorithm is validated for the first time using experimental data with two different anomalies.

The remainder of this paper is structured as follows. Section 2 introduces the NWATV regularizer and outlines the mathematical formulation of EIT. In Section 3, we perform both 2D and 3D numerical simulations along with water tank experiments under three-quarter-electrode and half-electrode-coverage configurations. Various metrics are employed to evaluate and compare the reconstruction performance of different regularization methods. Finally, Section 4 concludes with a brief summary and suggests potential directions for future research.

2. Inverse Problems in EIT and NWATV Regularizer

Starting from this section, we denote $\mathbf{r} = (x, y)$ and $\mathbf{r} = (x, y, z)$ for Ω , these being two- and three-dimensional objects, respectively. We denote $\gamma(\mathbf{r}, \omega) := \sigma(\mathbf{r}, \omega) + i\omega\epsilon(\mathbf{r}, \omega)$ for emphasizing the position and frequency dependency of γ . In this study we set $\omega/2\pi \leq 1$ MHz; hence, we will neglect the influence of the permittivity and are interested in only reconstructing σ .

The reconstruction of the conductivity distribution σ from the measured data U_i^j relies on the following reciprocity principle, which is derived by applying integration by parts to both sides of (1) [19]:

$$\begin{aligned}
 V_i^j[\sigma] &:= U_i^j - U_{i+1}^j \approx u^j|_{\mathcal{E}_i} - u^j|_{\mathcal{E}_{i+1}} \\
 &= \frac{1}{I} \int_{\Omega} \sigma(\mathbf{r}) \nabla u^i(\mathbf{r}) \cdot \nabla u^j(\mathbf{r}) d\mathbf{r} = \frac{1}{I} \int_{\Omega} \sigma(\mathbf{r}) \nabla u^j(\mathbf{r}) \cdot \nabla u^i(\mathbf{r}) d\mathbf{r} \\
 &= u^i|_{\mathcal{E}_j} - u^i|_{\mathcal{E}_{j+1}} \approx U_j^i - U_{j+1}^i =: V_j^i[\sigma].
 \end{aligned} \tag{2}$$

Here, u^i denotes the electric potential induced by current injection through the electrode pair $(\mathcal{E}_i, \mathcal{E}_{i+1})$. Note that the approximations in (2) hold only when the contact impedance is sufficiently small and can be neglected. Moreover, in (2), the indices i and j are exchanged through a simple application of the commutative law for the dot product of two vectors. Consequently, measurements associated with driving electrodes are considered unreliable, leaving $E(E-3)$ usable measurements for conductivity reconstruction.

To reconstruct the conductivity σ from the relation (2), we first observe that both potentials u^i and u^j themselves depend on the unknown σ , rendering the relationship between σ and the measurements V_i^j inherently nonlinear. To address this, we linearize (2) by replacing u^i and u^j with the corresponding potentials u_0^i and u_0^j , which are computed from (1) using a constant, homogeneous background conductivity σ_0 in Ω . This leads to the approximation

$$\int_{\Omega} \sigma(\mathbf{r}) \nabla u_0^i \cdot \nabla u_0^j d\mathbf{r} \approx IV_i^j[\sigma]. \tag{3}$$

For the reference conductivity σ_0 , the same relation holds exactly:

$$\int_{\Omega} \sigma_0(\mathbf{r}) \nabla u_0^i \cdot \nabla u_0^j d\mathbf{r} = IV_i^j[\sigma_0].$$

Subtracting these two relations yields the linearized equation

$$\int_{\Omega} \delta\sigma(\mathbf{r}) \nabla u_0^i \cdot \nabla u_0^j d\mathbf{r} = I\delta V_i^j, \tag{4}$$

where $\delta\sigma(\mathbf{r}) = \sigma(\mathbf{r}) - \sigma_0(\mathbf{r})$ and $\delta V_i^j = V_i^j[\sigma] - V_i^j[\sigma_0]$. The conductivity is then reconstructed as $\sigma = \sigma_0 + \delta\sigma$, with $\delta\sigma$ obtained by solving (4). In practice, we employ (4) rather than (3) for the reconstruction, because working with the difference δV_i^j helps mitigate modeling errors present in the measured data.

For two-dimensional EIT reconstruction, in order to solve (4), we discretize Ω into N triangular elements $T_1, T_2, \dots, T_N$, that is, $\Omega = \bigcup_{k=1}^N T_k$. Within each element, the conductivity $\sigma|_{T_k}$ is approximated by a constant value σ_k . This discretization allows σ to be represented by a vector $\boldsymbol{\sigma} := (\sigma_1, \sigma_2, \dots, \sigma_N)^T$, and hence $\delta\boldsymbol{\sigma} := (\delta\sigma_1, \delta\sigma_2, \dots, \delta\sigma_N)^T$. Under this discretization, (4) can be discretized into the following linear system:

$$\mathbf{S}\delta\boldsymbol{\sigma} = \delta\mathbf{V}.$$

Here, $\delta\mathbf{V} = [\delta V_1^3, \delta V_1^4, \dots, \delta V_1^{E-1}, \delta V_2^4, \dots, \delta V_E^{E-2}]^T$ is an $E(E-3)$ vector, and $\mathbf{S}$ is an $E(E-3) \times N$ sensitivity matrix defined as follows:

$$\mathbf{S} = \begin{bmatrix} \cdots & \begin{matrix} | \\ \mathbf{S}_k \\ | \end{matrix} & \cdots \end{bmatrix}, \quad \text{with } \mathbf{S}_k := \frac{1}{I} \int_{T_k} \mathbf{s}(\mathbf{r}) d\mathbf{r},$$

where $\mathbf{s}(\mathbf{r}) := [s^{1,3}(\mathbf{r}), s^{1,4}(\mathbf{r}), \dots, s^{E,(E-2)}(\mathbf{r})]^T$ with $s^{i,j}(\mathbf{r}) = \nabla u_0^i(\mathbf{r}) \cdot \nabla u_0^j(\mathbf{r})$.

The conductivity perturbation is then obtained by solving the following least-squares minimization problem:

$$\delta\sigma^* = \arg \min_{\delta\sigma} \|\mathbf{S}\delta\sigma - \delta\mathbf{V}\|_{l_2}^2, \quad (5)$$

where $\|\cdot\|_{l_2}$ is the Euclidean norm in $\mathbb{R}^{E(E-3)}$.

It is obvious that the solution of (5) is

$$\delta\sigma^* = (\mathbf{S}^T\mathbf{S})^{-1}\mathbf{S}^T\delta\mathbf{V}. \quad (6)$$

However, the matrix $\mathbf{S}^T\mathbf{S}$ is typically ill-conditioned [14], rendering the direct use of Formula (6) highly sensitive to electrode positioning and modelling errors. This numerical instability is exacerbated in the case of partial boundary measurements. Moreover, due to the regularizing nature of the forward Equation (1), the data $\delta\mathbf{V}$ are inherently insensitive to local conductivity variations. Therefore, to obtain a stable and reliable reconstruction, it is necessary to introduce prior information (regularization) into the solution process.

In this work, we employ the nonlinear weighted anisotropic total variation (NWATV) regularization as a prior. Specifically, alongside the least-squares formulation (5), we introduce the NWATV penalty term $\|\mathbf{p} \cdot \mathbf{D}\delta\sigma\|_{l_1}$, which promotes sparsity of $\mathbf{D}\delta\sigma$ while strongly preserving edges in $\delta\sigma$. Consequently, (5) is reformulated as

$$\delta\sigma_\lambda^* = \arg \min_{\delta\sigma} \left\{ \frac{1}{2} \|\mathbf{S}\delta\sigma - \delta\mathbf{V}\|_{l_2}^2 + \lambda \|\mathbf{p} \cdot \mathbf{D}\delta\sigma\|_{l_1} \right\}, \quad (7)$$

where $\lambda > 0$ is a regularization parameter that balances the fidelity term (the first term on the right-hand side) and the NWATV penalty $\|\mathbf{p} \cdot \mathbf{D}\delta\sigma\|_{l_1}$. Here, $\mathbf{D} = (\mathbf{D}_x; \mathbf{D}_y)$ denotes the discrete gradient operator, and the weight vector $\mathbf{p}$ is defined as

$$\mathbf{p} = [\zeta(\delta\sigma); \zeta(\delta\sigma)], \quad \zeta(\delta\sigma) = 1/(\|\mathbf{D}\delta\sigma\|^2 + \delta)$$

with $\delta > 0$ being a small constant preventing division by zero.

To solve the optimization problem (7), we adopt an alternating direction method of multipliers (ADMMs) framework. Specifically, we introduce an auxiliary variable $\mathbf{z} = \mathbf{D}\delta\sigma$ and reformulate the problem via the following augmented Lagrangian:

$$\mathcal{L}_\rho(\delta\sigma, \mathbf{z}, \mathbf{p}; \mathbf{y}) = \frac{1}{2} \|\mathbf{S}\delta\sigma - \delta\mathbf{V}\|_{l_2}^2 + \lambda \|\mathbf{p} \cdot \mathbf{z}\|_{l_1} + \frac{\rho}{2} \|\mathbf{D}\delta\sigma - \mathbf{z}\|_{l_2}^2 + \mathbf{y} \cdot (\mathbf{D}\delta\sigma - \mathbf{z}),$$

where $\rho > 0$ is a penalty parameter and $\mathbf{y}$ is the Lagrange multiplier. This transforms the original problem into a separable constrained optimization form, which can be solved iteratively by successively updating $\delta\sigma$, $\mathbf{z}$, $\mathbf{p}$, and $\mathbf{y}$ as follows:

$$\delta\sigma_{n+1} = \arg \min_{\delta\sigma} \mathcal{L}_\rho(\delta\sigma_n, \mathbf{z}_n, \mathbf{p}_n; \mathbf{y}_n); \quad (8)$$

$$\mathbf{z}_{n+1} = \arg \min_{\mathbf{z}} \mathcal{L}_\rho(\delta\sigma_{n+1}, \mathbf{z}, \mathbf{p}_n; \mathbf{y}_n); \quad (9)$$

$$\mathbf{p}_{n+1} = (\zeta(\delta\sigma_{n+1}); \zeta(\delta\sigma_{n+1})); \quad (10)$$

$$\mathbf{y}_{n+1} = \mathbf{y}_n + \rho(\mathbf{D}\delta\sigma_{n+1} - \mathbf{z}_{n+1}). \quad (11)$$

For practical implementation, the subproblems (8) and (9) admit the following closed-form solutions:

$$\delta\sigma_{n+1} = \left(\frac{1}{\rho} \mathbf{S}^T\mathbf{S} + \mathbf{D}^T\mathbf{D} \right)^{-1} \left(\frac{1}{\rho} \mathbf{S}^T\delta\mathbf{V} + \mathbf{D}^T\mathbf{z}_n - \mathbf{D}^T\frac{\mathbf{y}_n}{\rho} \right). \quad (12)$$

and

$$\mathbf{z}_{n+1}[k] = h_{\frac{\lambda}{\rho} \mathbf{p}_n[k]} \left(\mathbf{D} \delta \sigma_{n+1}[k] + \frac{\mathbf{y}_n[k]}{\rho} \right), \quad k = 1, 2, \dots, 2N, \quad (13)$$

where the shrinkage operator $h_g(x)$ is defined as

$$h_g(x) = \begin{cases} x - g \cdot \text{sgn}(x), & |x| > g, \\ 0, & |x| \leq g, \end{cases}$$

for any threshold $g \geq 0$. The detailed derivation of (12) and (13) can be found in [13].

We conclude this section by summarizing the above procedures in the following Algorithm 1 in the form of pseudocode. Here, we denote by $\mathbf{1}$ and $\mathbf{0}$ the vectors of all ones and all zeros, respectively.

Algorithm 1 NWATV based EIT reconstruction with partial boundary measurement

Input: Sensitivity matrix $\mathbf{S}$, observed data $\delta \mathbf{V}$.

Parameters: ρ, λ, δ , an error tolerance η and the maximum iteration number $M \in \mathbb{Z}^+$

Output: Reconstructed conductivity difference $\delta \sigma$

```

1: Initialize:  $\mathbf{z}_0 = \mathbf{0}$ ,  $\mathbf{p}_0 = (\mathbf{1}, \mathbf{1})$ ,  $\mathbf{y}_0 = \mathbf{0}$ .
2: for  $n = 0$  to  $M - 1$  do
3:   Update  $\delta \sigma_{n+1}$  by (12).
4:   Update  $\mathbf{z}_{n+1}$  by (13).
5:   Update  $\mathbf{p}_{n+1}$  by (10).
6:   Update  $\mathbf{y}_{n+1}$  by (11).
7:   Update  $\sigma_{n+1}$  by setting  $\sigma_{n+1} = \sigma_0 + \delta \sigma_{(n+1)}$ .
8:   if  $\|\sigma_{n+1} - \sigma_n\|_{l_2} < \eta$  then
9:     break
10:  end if
11: end for
```

3. Numerical and Water Tank Experiments

In this section, we perform numerical and water tank experiments to validate the proposed reconstruction Algorithm 1. In all experiments, 16 electrodes are attached equidistantly along the boundary, and adjacent current patterns are used for both current injection and voltage measurement. The performance of our method is compared with that of several conventional regularization approaches, including the fidelity-embedded regularizer (FER), the standard total variation (TV) regularizer, and the first-order TV regularizer. For quantitative evaluation, we compute the peak signal-to-noise ratio (PSNR) and the relative error (RE), and record the CPU time. At the n -th iteration step, $\text{RE}(n)$ and $\text{PSNR}(n)$ are defined as

$$\text{RE}(n) = \frac{\|\sigma_n - \sigma\|_{l_2}}{\|\sigma\|_{l_2}}, \quad (14)$$

and

$$\text{PSNR}(n) = 10 \log_{10} \frac{\max(\sigma_n \odot \sigma_n)}{\text{MSE}(n)}, \quad (15)$$

where $\cdot \odot \cdot$ denotes the componentwise multiplication and σ represents the true conductivity. The mean square error (MSE) between the estimated conductivity σ_n and σ is defined as $\text{MSE}(n) = \frac{1}{N} \sum_{i=1}^N [\sigma_n(i) - \sigma(i)]^2$. The CPU time (in seconds) is recorded using MATLAB's (version 2023a) tic and toc commands while running our code.

For the water tank experiments, where the true conductivity is unknown, we replace with the reconstruction σ_{TV} obtained using standard TV regularization as a reference.

3.1. Numerical Experiments

We adopt the same numerical phantom as in [13]. For completeness, the phantom is described as follows. A circular domain of radius 0.1 m is centered at the original point $O(0,0)$. Inside this disk, two ellipses are placed to mimic human lungs. The ellipse centers are fixed at $(0.04, -0.01)$ and $(-0.04, -0.01)$, respectively. To simulate a respiratory cycle, ten different configurations are generated by progressively enlarging the ellipses. For the k -th model ($k = 1, 2, \dots, 10$), the major axis vector for the left ellipse is $a_l = (0.012 + 0.001k, 0.024 + 0.002k)$ and its minor axis vector is $b_l = (-0.012 - 0.001k, 0.006 + 0.0005k)$; the major axis vector for the right ellipse is $a_r = (-0.012 - 0.001k, 0.024 + 0.002k)$ and its minor axis vector is $b_r = (0.012 + 0.001k, 0.006 + 0.0005k)$. The electrical conductivity of the disk and the internal ellipses are set to 1.0 S/m and 1.1 S/m, respectively. This model is shown in Figure 2a. To avoid ‘inverse crime’ [20], as shown in Figure 2a, in the process of solving the forward model, the corresponding meshes are set to be 2732 triangular elements and 1431 nodes, while in the process of reconstruction, we employ a finite element mesh with 1024 triangular elements and 545 nodes, as shown in Figure 2b.

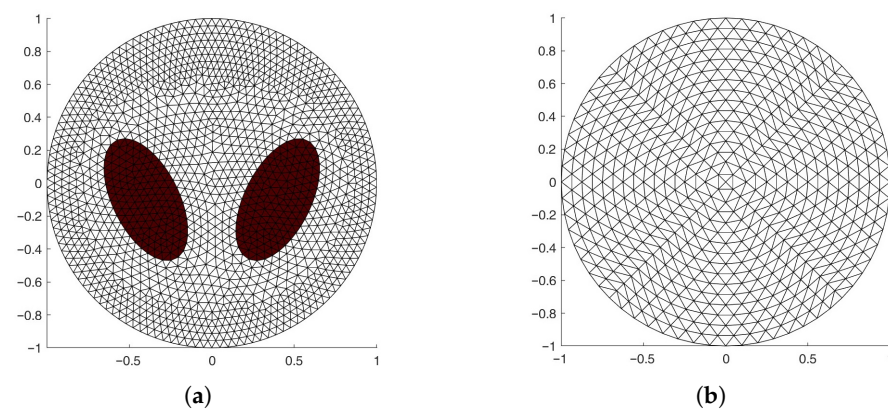


Figure 2. (a) Two-dimensional numerical model and its conductivity distribution. (b) Nodes and elements used in the inversion process. In (a), the higher conductivity (1.1 S/m), in contrast to the background value of 1 S/m, is represented by the color red.

The reconstruction results for the two electrode configurations—three-quarter circumferential arrays (Figure 3) and half-circumferential arrays (Figure 4)—are presented separately. The comparisons of REs and PSNRs for the reconstructions are, respectively, provided in Tables 1 and 2. The comparisons of CPU times for the reconstruction are respectively depicted in Tables 3 and 4. Each configuration is tested with four regularization methods: fidelity-embedded regularization (FER), standard total variation (TV), first-order TV, and the proposed NWATV. To be precise, the parameters we chose for the NWATV method are as follows: under the three-quarter circumferential electrode layout, $\lambda = 5 \times 10^{-18}$, $\rho = 5 \times 10^{-13}$, $\delta = 0.01$, while under the half-circumference configuration, $\lambda = 1 \times 10^{-19}$, $\rho = 3 \times 10^{-15}$, $\delta = 0.01$. We have explanations for the smallness of λ and ρ in [13]. All regularization parameters were manually tuned to their optimal values. For FER, only the limiting case with the regularization parameter set to infinity is considered. Except for FER, all iterative methods were run for 20 steps.

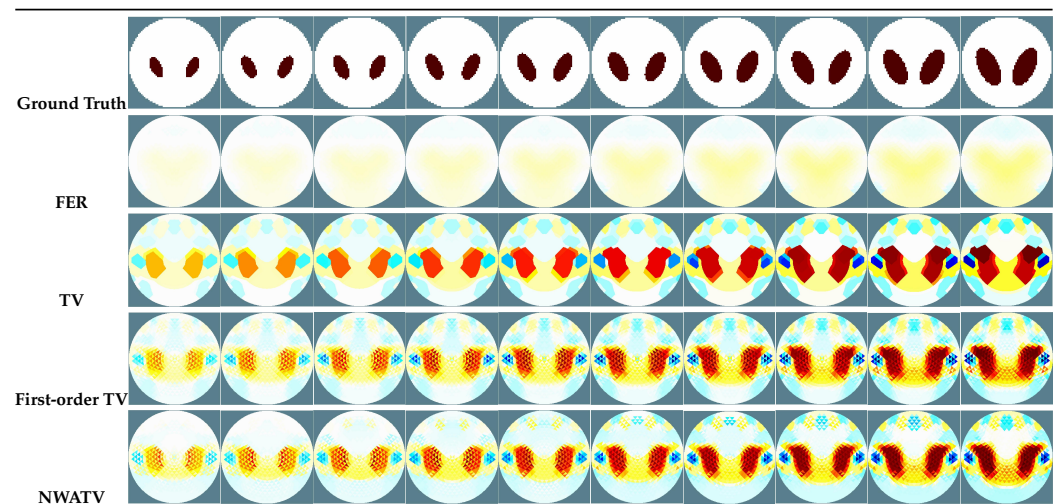


Figure 3. The reconstruction results of the three-quarter circular electrode layout model using four different regularization methods. The first line describes the true conductivity distribution, and the second to fifth lines show the reconstruction results using the FER, TV, first-order TV, and NWATV methods, respectively.



Figure 4. The reconstruction results of the half-circular electrode layout model using four different regularization methods. The First line describes the real conductivity distribution, and the second to fifth lines show the reconstruction results using the FER, TV, first-order TV, and NWATV methods.

As shown in Figure 3, the three-quarter circumferential electrode layout leads to a noticeable decline in reconstruction quality across all methods. The most prominent artifacts and distortions appear in the regions around the electrode-devoid sector. Despite this general degradation, the robustness of the regularization methods varies under such partial data conditions. The NWATV method demonstrates the best performance in preserving the elliptical shapes of the targets and minimizing overall distortion. In regions without electrode coverage, artifacts are effectively constrained and do not propagate extensively. This indicates that NWATV's nonlinear weighting mechanism can utilize limited data more effectively, suppressing artifacts while maintaining structural integrity. Consequently, NWATV exhibits superior stability, which is advantageous in clinically realistic, non-ideal measurement scenarios.

Table 1. The REs and PSNRs for $k = 1, 2, \dots, 10$ in the numerical experiments under a three-quarter electrode layout.

	TV	First-Order TV	NWATV
RE			
k = 1	0.0379	0.0378	0.0374
k = 2	0.0384	0.0381	0.0378
k = 3	0.0390	0.0383	0.0381
k = 4	0.0395	0.0386	0.0384
k = 5	0.0400	0.0398	0.0393
k = 6	0.0405	0.0400	0.0395
k = 7	0.0408	0.0401	0.0397
k = 8	0.0412	0.0402	0.0399
k = 9	0.0422	0.0402	0.0400
k = 10	0.0439	0.0403	0.0401
PSNR			
k = 1	19.8677	20.0031	20.8312
k = 2	19.7177	20.0886	20.8937
k = 3	19.5781	20.1939	20.9155
k = 4	19.4965	20.3347	20.9337
k = 5	19.1942	19.2315	19.7585
k = 6	19.1640	19.3951	19.9181
k = 7	19.1487	19.5776	20.0533
k = 8	19.1625	19.7735	20.2042
k = 9	18.9682	19.9457	20.3471
k = 10	18.5993	20.0909	20.4748

Table 2. The REs and PSNRs for $k = 1, 2, \dots, 10$ in the numerical experiments under a half-electrode layout.

	TV	First-Order TV	NWATV
RE			
k = 1	0.0409	0.0407	0.0406
k = 2	0.0418	0.0417	0.0414
k = 3	0.0417	0.0416	0.0415
k = 4	0.0428	0.0426	0.0423
k = 5	0.0444	0.0434	0.0431
k = 6	0.0462	0.0448	0.0442
k = 7	0.0484	0.0467	0.0450
k = 8	0.0509	0.0458	0.0453
k = 9	0.0533	0.0466	0.0462
k = 10	0.0550	0.0519	0.0473
PSNR			
k = 1	16.3362	16.6553	16.6624
k = 2	16.8025	17.3216	17.3793
k = 3	16.5265	17.0412	17.0984
k = 4	16.5255	16.8413	16.8430
k = 5	16.3424	16.5981	16.6574
k = 6	16.2573	16.4383	16.5063
k = 7	15.9609	16.1523	16.4725
k = 8	15.7371	16.0673	16.1575
k = 9	15.5774	15.7458	15.8251
k = 10	14.4221	15.5472	15.6171

When the electrode layout is further reduced to the more challenging half-circumference configuration, the quality of all reconstructed images deteriorates even more markedly, as seen in Figure 4. A critical limitation emerges: all algorithms fail to resolve the two distinct lung objects. Due to the complete absence of useful data from the entire lower region, each method incorrectly merges the two ellipses into a single, downward-curving arc. The lower half of the image is dominated by severe artifacts, which effectively compress the diagnostically useful region to the upper portion and strip the results of meaningful physiological structural information.

Table 3. The CPU time (in ‘s’) under different regularization methods in the numerical experiment with a three-quarter-electrode layout.

	FER	TV	First-Order TV	NWATV
k = 1	0.0128	1.3087	0.4330	0.4207
k = 2	0.0158	1.2791	0.4312	0.4163
k = 3	0.0165	1.4092	0.4306	0.4209
k = 4	0.0146	1.3787	0.4337	0.4237
k = 5	0.0128	1.2092	0.4927	0.4501
k = 6	0.0159	1.4388	0.4369	0.4227
k = 7	0.0134	1.2790	0.4305	0.4219
k = 8	0.0164	1.4145	0.3984	0.3753
k = 9	0.0141	1.3290	0.3803	0.3653
k = 10	0.0122	1.1894	0.3997	0.3849

Table 4. The CPU time (in ‘s’) under different regularization methods in the numerical experiment with a half-electrode layout.

	FER	TV	First-Order TV	NWATV
k = 1	0.0130	1.1793	0.4062	0.3622
k = 2	0.0127	1.3591	0.4163	0.4141
k = 3	0.0136	1.3190	0.4220	0.4011
k = 4	0.0177	1.2790	0.3805	0.3661
k = 5	0.0164	1.3394	0.3688	0.3726
k = 6	0.0267	1.4061	0.4849	0.4771
k = 7	0.0143	1.2858	0.3995	0.3654
k = 8	0.0156	1.3958	0.4556	0.3605
k = 9	0.0143	1.3588	0.4210	0.4037
k = 10	0.0124	1.3711	0.4193	0.4037

We compare the reconstruction performance of different regularization methods at stage $k = 8$, as summarized in Figure 5. As the electrode coverage is reduced from the full-circumference to three-quarter and, finally, half-coverage configuration, all methods exhibit a systematic decline in performance, characterized by increasing geometric distortion and boundary blurring. Under reduced electrode layouts, the FER method yields poorly defined reconstructions with substantial loss of detail. While the TV and first-order TV methods retain certain structural information, they introduce noticeable pseudo edges, often spuriously merging adjacent targets. In contrast, the NWATV method maintains superior edge preservation across all configurations. Its nonlinear weighted–anisotropic mechanism effectively leverages the limited boundary data to yield reconstructions that most closely approximate the true conductivity distribution.

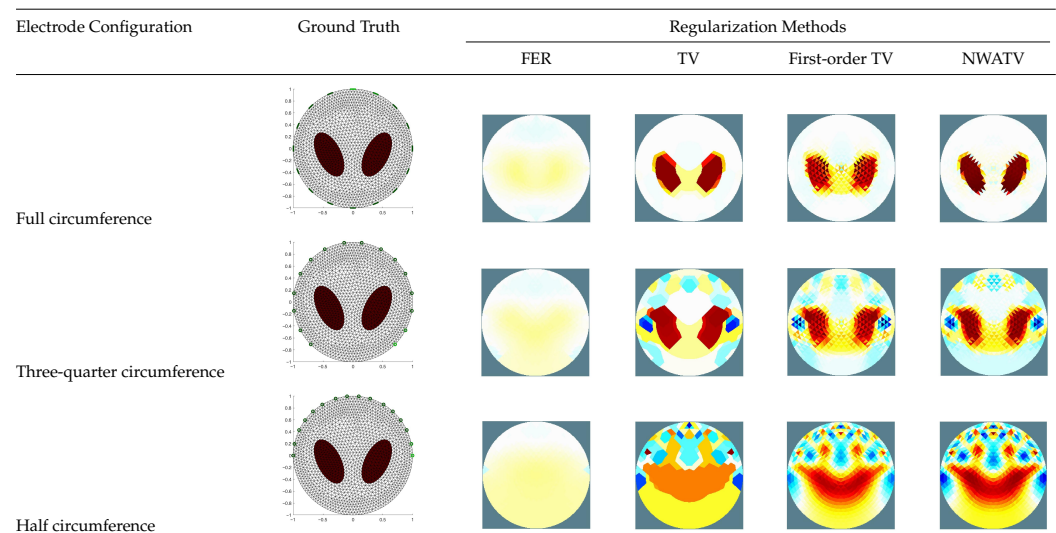


Figure 5. Comparison of the reconstruction results of two-dimensional models in different regularization methods under different electrode configurations when $k = 8$.

For further demonstrate the better performance of the NWATV regularizer, we compare the PSNR, RE and CPU time. Figure 6a,b illustrate the behavior of $PSNR(n)$ and $RE(n)$ as n increases for the scenario in which the electrodes covers three-quarters of the boundary $\partial\Omega$, while Figure 6c,d show the scenario in which the electrodes cover only half of the object boundary. For the three-quarter electrode layouts in Figure 6a,b, though the blue curve achieves its maximum PSNR and minimum RE values at the seventh iteration, its performance deteriorates rapidly thereafter, exhibiting significant instability. In contrast, the red curve achieves optimal performance during iteration, attaining the lowest RE and highest PSNR values. Moreover, it remains relatively stable after reaching its performance peak. For the half-circumference layouts in Figure 6c,d, PSNR of the blue curve collapses sharply after reaching its best value at iteration 9. While the performance of the red curve also fails to improve with further iteration, its degradation process is the most gradual, maintaining the optimal performance overall. This shows that algorithms struggle to improve the solution under severely deficient boundary data. Regarding CPU time, when the electrodes cover three-quarters of the boundary $\partial\Omega$, the recorded times for FER, TV, first-order TV, and NWATV are 0.0164 s, 1.4145 s, 0.3984 s and 0.3753 s, respectively. In the half-circumference case, the corresponding CPU times are 0.0156 s, 1.3958 s, 0.4556 s and 0.3605 s, respectively.

In conclusion, NWATV achieves the best performance while requiring relatively less computational time.

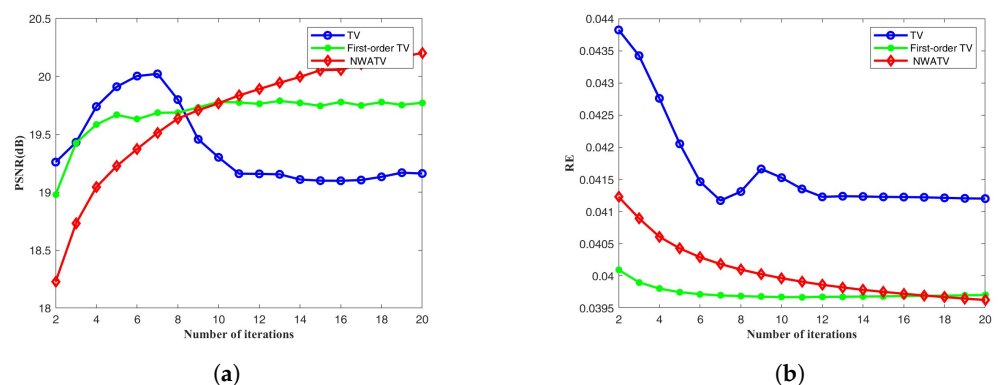


Figure 6. Cont.

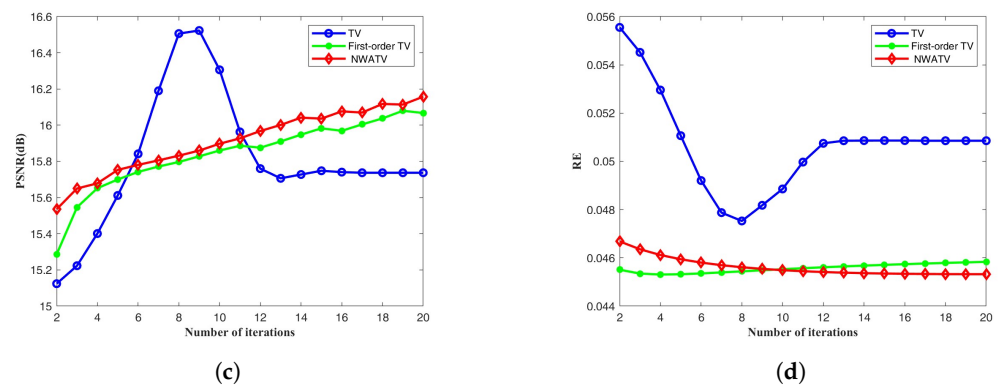


Figure 6. PSNR and RE curves under different electrode configurations. (a) PSNR under three-quarter layout, (b) RE under three-quarter layout, (c) PSNR under half-circumference layout, and (d) RE under half-circumference layout.

3.2. Water Tank Experiment

To validate the performance of NWATV in electrical impedance tomography (EIT) reconstruction using partial boundary measurements, we perform two water tank experiments with a Sciospec 16-channel EIT system. A cylindrical tank with a radius of 10 cm is filled with tap water as the homogeneous background medium. Sixteen electrodes are arranged in different configurations, and an adjacent stimulation-measurement protocol is applied with an injection current of 1 mA. The data acquisition rate is approximately 20 frames per second. Boundary voltage measurements taken without any target present yield an estimated reference conductivity of water, $\sigma_0 = 0.003$ S/m. The inversion model is discretized using a finite element mesh consisting of 841 nodes and 1600 triangular elements.

In the experiments, carrot, cucumber, and beancurd are individually placed within the tank as imaging targets. These targets are then arranged in pairwise combinations to simulate multiple scenarios. Boundary voltage data are acquired for all experimental configurations. The image reconstruction is performed using FER, TV, first-order TV, and NWATV regularization methods, with the iterative process set to 20 steps.

Under the three-quarter-circumferential electrode configuration, the corresponding reconstruction results are displayed in the second to fourth rows in Figure 7. We can see that all regularization methods achieve the basic localization of single target, albeit with significant disparities in reconstruction quality. The FER method yields images with pronounced background noise and blurred target contours, providing only approximate position information. The TV method delineates the target shape more clearly, yet the edges exhibit serrated artifacts and slight geometric distortion. The first-order TV method offers higher computational efficiency. However, it causes directional distortion along target edges and generates structural artifacts in regions devoid of electrodes. In contrast, the NWATV produces reconstructed images with clearer target contours and well-preserved edges, while reducing the background artifacts significantly. This demonstrates its favorable practical utility.

The results for the multi-target scenarios are shown in the last three rows of Figure 7. The difficulty of reconstruction increases substantially. The FER method fails to distinguish the two adjacent targets, resulting in a blurred connected region. Though the TV method can indicate the presence of two targets, their boundaries remain blurred with severe shape distortion, and the inter-target distance appears compressed. The first-order TV method exhibits noticeable instability, manifesting as the spurious connections between targets and fusion phenomena at specific angles. The NWATV method maintains its distinguishing

advantage, and separates adjacent targets with clarity successfully. Moreover, it preserves the relative positions, and provides better edge preservation.

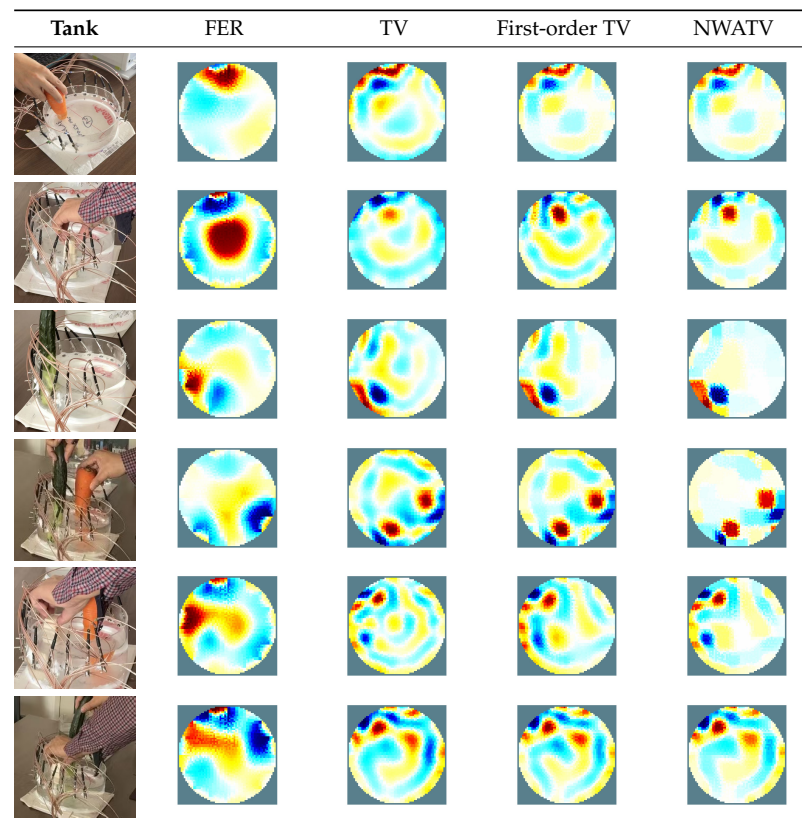


Figure 7. The reconstruction results of the water tank experiment by four different regularization methods under a three-quarter-circular-electrode layout.

The CPU times and RE(20)/PSNR(20) for each case are displayed in Table 5 and Table 6, respectively. As we can see from these quantitative results, NWATV behaves better than other regularization methods.

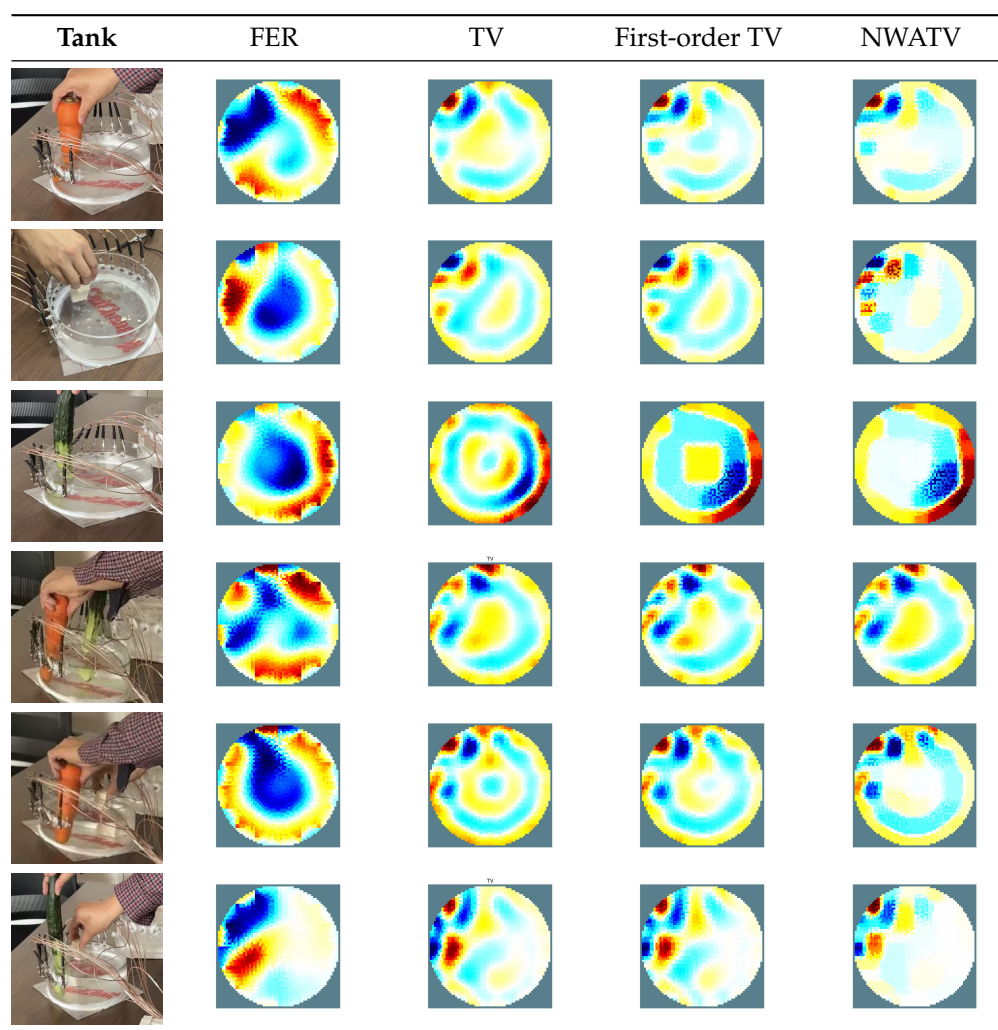
Table 5. The RE and PSNR of different models in the water tank experiment under a three-quarter-electrode layout.

	FER, $\lambda = \infty$	First-Order TV	NWATV
RE			
Carrot	0.09997	0.09913	0.09913
Beancurd	0.09999	0.09958	0.09859
Cucumber	0.09998	0.09997	0.09997
Carrot and Cucumber	0.09999	0.09995	0.09998
Carrot and Beancurd	0.09998	0.09979	0.09958
Beancurd and Cucumber	0.09995	0.09983	0.09984
PSNR			
Carrot	17.44	21.52	21.52
Beancurd	13.43	19.61	23.72
Cucumber	19.18	19.24	19.51
Carrot and Cucumber	14.26	16.01	15.42
Carrot and Beancurd	15.81	18.49	20.52
Beancurd and Cucumber	15.14	18.07	18.13

Table 6. The CPU time (in ‘s’) under different regularization methods in a water tank experiment with a three-quarter-electrode layout.

	FER	TV	First-Order TV	NWATV
Carrot	0.0408	3.0162	1.0720	1.0529
Beancurd	0.0423	3.0323	1.1013	1.0603
Cucumber	0.0365	3.2276	1.2702	1.2682
Carrot and Cucumber	0.0368	3.0252	1.1416	1.1394
Carrot and Beancurd	0.0353	3.0186	0.9802	0.9992
Beancurd and Cucumber	0.0453	3.0189	1.1447	1.1306

For the half-circumference-electrode configuration, the reconstruction results are displayed in Figure 8. According to these results, the reconstruction quality for a single target deteriorates generally. The FER method becomes nearly ineffective, as the target becomes indistinguishable from the background noise. The TV method can still localize the target region roughly, but the reconstructed shape suffers severe distortion with exacerbated edge serration. The first-order TV method causes more pronounced target distortion, exhibiting significant stretching along the direction lacking electrode coverage. However the NWATV method maintains the fundamental morphological features of the target.

**Figure 8.** The reconstruction results of the water tank experiment using four different regularization methods under a half-circular electrode layout.

Under this electrode configuration, multi-target reconstruction faces greater challenges. The FER method loses its practical value, failing to provide useful target information. Although the TV and first-order TV methods can indicate the presence of targets, they merge them into a single connected region erroneously. The NWATV method maintains the separation of the two targets. Though the image quality degrades somewhat, it avoids structural failure.

Quantitative comparisons shown in Tables 7 and 8 also indicate that NWATV behaves better than other regularization methods.

Table 7. The RE and PSNR of different models in the water tank experiment under a half-electrode layout.

	FER, $\lambda = \infty$	First-Order TV	NWATV
RE			
Carrot	0.09999	0.09971	0.09972
Beancurd	0.09998	0.09998	0.09994
Cucumber	0.10014	0.09937	0.09955
Carrot and Cucumber	0.09998	0.09818	0.09869
Carrot and Beancurd	0.10002	0.09945	0.09912
Beancurd and Cucumber	0.09999	0.09991	0.09976
PSNR			
Carrot	13.01	18.88	18.97
Beancurd	15.80	16.56	17.82
Cucumber	12.74	16.39	16.48
Carrot and Cucumber	13.83	21.55	19.98
Carrot and Beancurd	13.63	19.21	20.62
Beancurd and Cucumber	15.41	18.78	20.18

Table 8. The CPU time (in ‘s’) by different regularization methods in a water tank experiment with a half-electrode layout.

	FER	TV	First-Order TV	NWATV
Carrot	0.0382	3.0193	1.1434	1.1404
Beancurd	0.0557	3.0185	1.0441	1.0394
Cucumber	0.0475	3.1089	1.1542	1.1023
Carrot and Cucumber	0.0416	3.0207	1.0643	1.1094
Carrot and Beancurd	0.0383	3.1185	1.1584	1.1876
Beancurd and Cucumber	0.0608	3.2325	1.2849	1.2395

4. Concluding Remarks

This work examines the impact of partial boundary measurements on image reconstruction quality in electrical impedance tomography and provides a focused evaluation of four regularization methods under three-quarter-electrode and half-circumferential-electrode configurations. The results show a systematic decline in image quality across all reconstruction methods as electrode coverage is reduced. However, the NWATV method consistently demonstrates superior structural fidelity and robustness under every restricted measurement condition. This conclusion is supported by both quantitative metrics, including PSNR and RE, and qualitative analysis. Specifically, whether a single object or two objects are placed inside the water tank, the TV regularization method exhibits a noticeable staircasing effect, while the reconstruction using first-order TV shows distortion along the coordinate axes. In contrast, the NWATV regularization method demonstrates excellent stability even under severely insufficient boundary data. Despite its increased complexity compared to that of first-order TV regularization, NWATV shows a clear improvement

in distinguishing the two objects with fewer reconstruction artifacts, as evidenced by the second, fourth, fifth, and sixth experiments in Figures 7 and 8. In summary, this work verifies that the NWATV regularization method can mitigate reconstruction degradation caused by partial boundary electrode coverage effectively. It provides a stable and practical solution for clinical EIT monitoring with constrained electrode layouts. This finding holds significant implications for enhancing the clinical availability of EIT technology.

Future work will focus on further optimizing the computational efficiency of the NWATV regularization and exploring its integration with deep learning [21] to achieve faster and more accurate reconstruction.

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